Day 7 — CSS Fundamentals

Completed Assignment: HTML + CSS Programs

This PDF completes all five tasks in the Day 7 assignment: style a personal profile page, improve the student registration form, create a navigation bar, arrange cards with Flexbox, and make the pages responsive for mobile and desktop screens.

It also explains inline, internal, and external CSS; selectors and properties; colors; margin and padding; the box model; Flexbox; and media queries.

1. Introduction to CSS

CSS means Cascading Style Sheets. It controls the appearance and layout of HTML, including colors, fonts, spacing, borders, and responsive layouts.

Inline CSS: written in an element's style attribute, e.g. Hello .

Internal CSS: written inside a <style> element in the HTML head:

```
<style>
p { color: blue; }
</style>
```

External CSS: saved in a separate .css file and connected with <link rel="stylesheet" href="style.css">. External CSS is useful for sharing styles across pages. The tasks below use external CSS.

2. Task 1 — Style the Personal Profile Page

Create **profile.html** and **style.css** in the same folder.

File: profile.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Personal Profile</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="site-header">
    <div class="container">
      <h1>Shruti's Profile</h1>
      <p>Computer Engineering Graduate</p>
    </div>
  </header>

  <nav class="navbar">
    <div class="container nav-links">
      <a href="#about">About</a>
      <a href="#education">Education</a>
      <a href="#skills">Skills</a>
      <a href="#contact">Contact</a>
    </div>
  </nav>

  <main class="container">
    <section id="about" class="profile-card">
      <div class="profile-intro">
        <img src="profile.jpg" alt="Profile photo" class="profile-image">
        <div>
          <h2>About Me</h2>
          <p>Hello! I am a Computer Engineering graduate interested in learning new technologies and building useful web
applications.</p>
        </div>
      </div>
    </section>

    <section class="card-grid">
      <article id="education" class="info-card">
        <h2>Education</h2>
        <p><strong>Degree:</strong> B.E. Computer Engineering</p>
        <p><strong>Interest:</strong> Web technologies</p>
      </article>
      <article id="skills" class="info-card">
        <h2>Technical Skills</h2>
        <ul>
          <li>HTML</li><li>CSS</li><li>Java</li><li>MySQL</li>
        </ul>
      </article>
    </section>

    <section id="contact" class="contact-card">
      <h2>Contact</h2>
      <p>Email: <a href="mailto:example@gmail.com">example@gmail.com</a></p>
      <p>Replace sample contact details with your own.</p>
    </section>
  </main>

  <footer class="site-footer">
    <p>&copy; 2026 Shruti's Profile</p>
  </footer>
</body>
</html>
```

File: style.css

```
* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body {
  margin: 0;
  font-family: Arial, sans-serif;
  line-height: 1.6;
  color: #243247;
```

```
  background: #f3f6fb;
}
.container { width: min(1100px, 92%); margin: 0 auto; }
.site-header {
  background: linear-gradient(135deg, #17365d, #2874a6);
  color: white; padding: 28px 0; text-align: center;
}
.site-header h1 { margin: 0; }
.site-header p { margin: 5px 0 0; }
.navbar { background: #10263f; }
.nav-links {
  display: flex; justify-content: center; flex-wrap: wrap;
  gap: 12px; padding: 12px 0;
}
.nav-links a {
  color: white; text-decoration: none;
  padding: 8px 14px; border-radius: 6px;
}
.nav-links a:hover, .nav-links a:focus { background: #2874a6; }
main.container { padding-top: 24px; padding-bottom: 24px; }
.profile-card, .info-card, .contact-card {
  background: white; padding: 22px; margin-bottom: 20px;
  border-radius: 12px; box-shadow: 0 3px 12px rgba(0,0,0,.08);
}
.profile-intro { display: flex; align-items: center; gap: 22px; }
.profile-image {
  width: 140px; height: 140px; object-fit: cover;
  border-radius: 50%; background: #e1e8f0;
}
.card-grid { display: flex; gap: 20px; align-items: stretch; }
.info-card { flex: 1; }
.info-card h2, .contact-card h2, .profile-card h2 { color: #17365d; }
.site-footer { text-align: center; padding: 16px; color: white; background: #10263f; }
@media (max-width: 600px) {
  .profile-intro { flex-direction: column; text-align: center; }
  .card-grid { flex-direction: column; gap: 0; }
  .nav-links { gap: 4px; }
  .nav-links a { padding: 8px; }
  .profile-card, .info-card, .contact-card { padding: 16px; }
}
```

Expected output: A colored header and navigation bar, a profile card, education and skills cards side by side on wider screens, and stacked cards on mobile. Put an image named profile.jpg in the same folder for the profile photo.

3. Task 2 — Improve the Student Registration Form

Create **registration.html** and **form.css** in the same folder.

File: registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration</title>
  <link rel="stylesheet" href="form.css">
</head>
<body>
  <header class="form-header">
    <h1>Student Registration</h1>
    <p>Enter your details to complete registration.</p>
  </header>

  <main class="form-container">
    <form action="#" method="post">
      <fieldset>
        <legend>Personal Information</legend>
        <div class="form-group">
          <label for="name">Full Name</label>
          <input type="text" id="name" name="name" placeholder="Enter full name" required>
        </div>
        <div class="form-group">
          <label for="email">Email Address</label>
          <input type="email" id="email" name="email" placeholder="you@example.com" required>
        </div>
        <div class="form-group">
          <label for="phone">Phone Number</label>
          <input type="tel" id="phone" name="phone" placeholder="Enter phone number" required>
        </div>
        <div class="form-group">
          <label for="dob">Date of Birth</label>
          <input type="date" id="dob" name="dob" required>
        </div>
        <div class="form-group">
          <span class="field-label">Gender</span>
          <div class="options">
            <label><input type="radio" name="gender" value="Male"> Male</label>
            <label><input type="radio" name="gender" value="Female"> Female</label>
            <label><input type="radio" name="gender" value="Other"> Other</label>
          </div>
        </div>
      </fieldset>

      <fieldset>
        <legend>Educational Information</legend>
        <div class="form-group">
          <label for="course">Course</label>
          <select id="course" name="course" required>
            <option value="">Choose a course</option>
            <option value="BE">B.E.</option>
            <option value="BTech">B.Tech</option>
            <option value="BCA">BCA</option>
            <option value="MCA">MCA</option>
          </select>
        </div>
        <div class="form-group">
          <span class="field-label">Skills</span>
          <div class="options">
            <label><input type="checkbox" name="skills" value="HTML"> HTML</label>
            <label><input type="checkbox" name="skills" value="CSS"> CSS</label>
            <label><input type="checkbox" name="skills" value="Java"> Java</label>
            <label><input type="checkbox" name="skills" value="MySQL"> MySQL</label>
          </div>
        </div>
        <div class="form-group">
          <label for="address">Address</label>
          <textarea id="address" name="address" rows="4" placeholder="Enter address"></textarea>
        </div>
      </fieldset>

      <div class="form-actions">
        <button type="submit">Register</button>
        <button type="reset" class="reset-button">Reset</button>
      </div>
    </form>
    <p class="note">Demo only: connect a backend to process and save submissions.</p>
  </main>
</body>
</html>
```

File: form.css

```
* { box-sizing: border-box; }
body {
  margin: 0; font-family: Arial, sans-serif;
  color: #243247; background: #f3f6fb; line-height: 1.5;
}
.form-header {
  text-align: center; color: white; background: #17365d;
  padding: 26px 16px;
}
.form-header h1 { margin: 0 0 6px; }
```

```
.form-header p { margin: 0; }
.form-container {
  width: min(700px, 92%); margin: 26px auto; padding: 24px;
  background: white; border-radius: 12px;
  box-shadow: 0 3px 12px rgba(0,0,0,.08);
}
fieldset {
  border: 1px solid #cbd5e1; border-radius: 8px;
  padding: 18px; margin: 0 0 20px;
}
legend { font-weight: bold; color: #17365d; padding: 0 6px; }
.form-group { margin-bottom: 16px; }
.form-group > label, .field-label {
  display: block; font-weight: bold; margin-bottom: 6px;
}
input[type="text"], input[type="email"], input[type="tel"],
input[type="date"], select, textarea {
  width: 100%; padding: 11px; border: 1px solid #b8c4d2;
  border-radius: 6px; font: inherit;
}
input:focus, select:focus, textarea:focus {
  outline: 2px solid #2874a6; border-color: #2874a6;
}
.options { display: flex; flex-wrap: wrap; gap: 12px; }
.options label { font-weight: normal; }
.form-actions { display: flex; gap: 12px; flex-wrap: wrap; }
button {
  border: 0; border-radius: 6px; padding: 11px 20px;
  font: inherit; cursor: pointer; color: white; background: #176b45;
}
button:hover { opacity: .9; }
button.reset-button { background: #5b6573; }
.note { color: #596579; font-size: .9rem; margin-bottom: 0; }
@media (max-width: 600px) {
  .form-container { width: 94%; padding: 16px; margin: 14px auto; }
  fieldset { padding: 12px; }
  .options { flex-direction: column; gap: 8px; }
  .form-actions { flex-direction: column; }
  button { width: 100%; }
}
```

Expected output: A centered registration form with a dark header, grouped fields, styled inputs and buttons, and a narrow-screen layout that fits mobile devices.

Important: The form uses action="#" as a placeholder. HTML/CSS alone does not save records; connect a backend or form service to process submissions.

4. Task 3 — Create a Navigation Bar

The profile page already includes a responsive navigation bar. Here is a compact standalone version:

```
<nav class="navbar">
  <a href="#home">Home</a>
  <a href="#about">About</a>
  <a href="#services">Services</a>
  <a href="#contact">Contact</a>
</nav>
```

```
.navbar {
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
  gap: 10px;
  background: #10263f;
  padding: 12px;
}
.navbar a {
  color: white;
  text-decoration: none;
  padding: 8px 14px;
  border-radius: 6px;
}
.navbar a:hover, .navbar a:focus { background: #2874a6; }
@media (max-width: 600px) {
  .navbar { gap: 4px; }
  .navbar a { padding: 8px; }
}
```

Explanation: display: flex lays out links in a row; gap spaces them; flex-wrap: wrap lets links wrap when needed; and the media query adjusts spacing on small screens.

5. Task 4 — Arrange Cards Using Flexbox

Flexbox arranges elements in a row or column and helps align them.

HTML

```
<section class="card-grid">
  <article class="card">
    <h2>Education</h2><p>B.E. Computer Engineering</p>
  </article>
  <article class="card">
    <h2>Skills</h2><p>HTML, CSS, Java, MySQL</p>
  </article>
  <article class="card">
    <h2>Projects</h2><p>Web application projects</p>
  </article>
</section>
```

CSS

```
.card-grid {
  display: flex;
  gap: 20px;
  align-items: stretch;
}
.card {
  flex: 1;
  padding: 20px;
  background: white;
  border-radius: 10px;
  box-shadow: 0 3px 12px rgba(0,0,0,.08);
}
@media (max-width: 600px) {
  .card-grid { flex-direction: column; gap: 12px; }
}
```

Important Flexbox properties: display: flex activates Flexbox; flex-direction chooses row or column; justify-content distributes items along the main axis; align-items aligns items along the cross axis; gap adds spacing; and flex-wrap allows wrapping.

6. Task 5 — Responsive Design

Responsive design allows a webpage to adapt to mobile, tablet, and desktop screen sizes.

Viewport meta tag (place inside the HTML head):

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Flexible container:

```
.container {
  width: min(1100px, 92%);
  margin: 0 auto;
}
```

Media query for smaller screens:

```
@media (max-width: 600px) {
  .card-grid { flex-direction: column; }
}
```

When the viewport is 600 pixels wide or less, the media-query rules apply and the cards stack vertically. Test both a wide desktop window and a narrow mobile window.

7. CSS Concepts — Quick Reference

Concept	Meaning	Example
Selector	Chooses which HTML elements to style.	p { color: blue; }
Property	The setting being changed.	color
Value	The setting applied.	blue
Color/background	Controls text and background appearance.	background: #f3f6fb;
Margin	Space outside the border.	margin: 20px;
Padding	Space between content and border.	padding: 16px;
Box model	Content + padding + border + margin.	box-sizing: border-box;
Flexbox	Flexible item layout.	display: flex;
Media query	Applies CSS under conditions.	@media (max-width: 600px)

Box model: Each element has content inside padding, then a border, with margin outside the border. box-sizing: border-box makes the declared width include padding and border.

8. How to Run the Assignment

Create a folder named **Day7_CSS** and place these files inside it: profile.html, style.css, registration.html, and form.css. Keep each HTML file beside its matching CSS file. Open the HTML files in a browser.

If styles do not appear, check that the stylesheet filename in the HTML link tag exactly matches the actual CSS filename. For the profile image, put profile.jpg in the same folder or update the image path.

9. Short Interview Answers

What is CSS? CSS stands for Cascading Style Sheets and controls the presentation and layout of HTML.

Inline vs internal vs external CSS? Inline CSS is in an element's style attribute; internal CSS is in a style element in the HTML head; external CSS is in a separate .css file.

What is the box model? It describes an element as content, padding, border, and margin.

What is Flexbox? A CSS layout model for arranging and aligning elements in rows or columns.

What is responsive design? A design approach that adapts webpage layout to different screen sizes.

What is a media query? It applies CSS rules when a condition, such as viewport width, is met.

Final result: All five Day 7 tasks are covered: styled profile page, improved responsive registration form, navigation bar, Flexbox cards, and mobile/desktop responsiveness.